

Assignment 4

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Part 1

The estimated model we are using is:

$$\log(\text{inc}_i) = \beta_0 + \beta_1 \cdot \text{fsize}_i + \beta_2 \cdot \log(\text{tw}_i) + \beta_3 \cdot \text{age}_i + \beta_4 \cdot \text{age}_i^2 + \beta_5 \cdot \text{educ}_i + \beta_6 \cdot \text{marr}_i + u_i, \quad i \in \{1, \dots, N\}.$$

Exercise 1 Coefficient interpretation

(a) Interpretation of $\hat{\beta}_1(\text{fsize})$, $\hat{\beta}_2(\log(\text{tw}))$, and $\hat{\beta}_6(\text{marr})$

- $\hat{\beta}_1(\text{fsize})$ (Log-level): Ceteris paribus, one additional household member is associated with a change in household income of $(e^{\hat{\beta}_1} - 1) \times 100 = (e^{-0.0412} - 1) \times 100 \approx -4.04\%$. The economically correct interpretation formula is iii. $(e^{\hat{\beta}_j} - 1) \times 100$, because the dependent variable is in logs and fsize is in levels; the approximation ii. $\hat{\beta}_j \times 100 = -4.12\%$ is very close.
- $\hat{\beta}_2(\log(\text{tw}))$ (Log-log): Ceteris paribus, a 1% increase in total household wealth is associated with an increase in household income of approximately 0.2873%. The economically correct interpretation formula is i. $\hat{\beta}_j$, because both the dependent variable and the independent variable are in logs, so the coefficient is an elasticity.
- $\hat{\beta}_6(\text{marr})$ (Log-level dummy): Ceteris paribus, if the main earner is married, household income is associated with an exact difference of $(e^{\hat{\beta}_6} - 1) \times 100 = (e^{0.4217} - 1) \times 100 \approx 52.46\%$ relative to an unmarried main earner. The economically correct interpretation formula is iii. $(e^{\hat{\beta}_j} - 1) \times 100$, because marr is a dummy variable in a log-level model; the approximation ii. $\hat{\beta}_j \times 100 = 42.17\%$ differs meaningfully, so the exact value should be reported.

(b) age and age^2 ; marginal effects

1. The marginal effect of age on $\log(\text{inc}_i)$ is

$$\frac{\partial \log(\text{inc}_i)}{\partial \text{age}_i} = \hat{\beta}_3 + 2 \cdot \hat{\beta}_4 \text{age}_i.$$

At $\text{age} = 25$ this equals

$$0.0512 + 2 \cdot (-0.0006) \cdot 25 = 0.0212.$$

Therefore, *ceteris paribus*, one additional year of age at age 25 is associated with approximately 2.12% higher household income.

2. Solving for the turning point age^* :

$$0 = 0.0512 + 2 \cdot (-0.0006) \cdot age^*$$

$$age^* = \frac{-0.0512}{2 \cdot (-0.0006)} = 42.67.$$

Therefore, the fitted age-income profile increases until approximately age 42.7 and decreases afterwards, holding all other regressors constant.

Exercise 2 Model specification

$$\mathbb{E}[\check{\beta}_j] = \beta_j + \underbrace{\beta_{tw} \cdot \delta_j}_{\text{bias}}, \quad \delta_j \approx \frac{\text{Cov}(x_j, \text{twoearn})}{\text{Var}(x_j)}.$$

The direction of the bias on:

- $\check{\beta}_1(\text{fsize})$:
 1. $\beta_{tw} > 0$ because a second employed person in the household should increase household income.
 2. $\delta_1 > 0$ because larger households are more likely to include two employed persons.
 3. Since $\beta_{tw} \cdot \delta_1 > 0$, $\mathbb{E}[\check{\beta}_1] > \beta_1$, so the coefficient on fsize is biased upward.
- $\check{\beta}_6(\text{marr})$:
 1. $\beta_{tw} > 0$ because a second employed household member should raise household income.
 2. $\delta_6 > 0$ because married main earners are more likely to live in a household with a second employed person.
 3. Since $\beta_{tw} \cdot \delta_6 > 0$, $\mathbb{E}[\check{\beta}_6] > \beta_6$, so the coefficient on marr is biased upward.

Exercise 3 Residual patterns and inference

(a) Gauss-Markov assumption

This pattern calls into question MLR.5: homoscedasticity, because instead of

$$\text{Var}(u_i \mid \text{fsize}_i, \log(\text{tw}_i), \text{age}_i, \text{age}_i^2, \text{educ}_i, \text{marr}_i) = \sigma^2$$

for all observations, the residual plot suggests that the conditional error variance is larger for households with higher education.

(b) Unbiasedness and BLUE

1. OLS is still unbiased if MLR.1-MLR.4 hold, because homoscedasticity is not required for unbiasedness.
2. OLS is no longer BLUE, because under heteroscedasticity OLS is no longer necessarily the estimator with the smallest variance among linear unbiased estimators.

(c) Sticking to homoscedasticity-based formula

If homoscedasticity-based standard errors are used, they can be wrong because they impose constant error variance. Therefore, the usual t -tests, F -tests, confidence intervals, and p -values may be unreliable. We should therefore use heteroscedasticity-robust standard errors for inference.

Exercise 4 Inference for household size

1. Hypotheses:

$$H_0 : \beta_1 \geq 0 \quad \text{vs} \quad H_1 : \beta_1 < 0, \quad \alpha = 1\%.$$

2. Test statistic:

$$t = \frac{\hat{\beta}_1 - 0}{se(\hat{\beta}_1)}.$$

3. Observed t -value: As $\hat{\beta}_1 = -0.0412$ and $se(\hat{\beta}_1) = 0.0201$,

$$t = \frac{-0.0412}{0.0201} = -2.0498.$$

4. Reference distribution under H_0 : Under the null hypothesis, the test statistic follows a t -distribution with

$$df = N - k - 1 = 534 - 6 - 1 = 527$$

degrees of freedom:

$$T \sim t_{527}.$$

Here, $k = 6$ is the number of regressors excluding the intercept.

5. Critical value at $\alpha = 1\%$:

Because this is a left-sided test, the critical value is the 1% quantile of the t_{527} distribution (from `qt(0.01, df = 527)`):

$$c = t_{0.01, 527} \approx -2.334.$$

6. p -value (from `pt(-2.0498, df = 527)`):

$$p = P(T_{527} \leq -2.0498) \approx 0.0204.$$

7. Decision: Since $-2.0498 > -2.334$ and $0.0204 > 0.01$, we do not reject H_0 at the 1% significance level. There is not enough statistical evidence at the 1% level to conclude that household size has a negative effect on household income.

Exercise 5 Testing age effects

1. Separate t-tests for age and age^2 are not the correct way to test whether age has any effect jointly, because the effect of age is represented by both coefficients together. Even if one coefficient were individually insignificant, the two variables could still be jointly significant.
2. The correct test is an F -test for joint significance:

$$H_0 : \beta_3 = 0 \text{ and } \beta_4 = 0$$

against

$$H_1 : \text{at least one of } \beta_3, \beta_4 \text{ is different from zero.}$$

3. The test statistic is

$$F = \frac{(SSR_r - SSR_{ur})/q}{SSR_{ur}/(N - k - 1)}.$$

Here, SSR_r is the sum of squared residuals from the restricted model, SSR_{ur} is the sum of squared residuals from the unrestricted model, $q = 2$ is the number of restrictions, $N = 534$ is the sample size, and $k = 6$ is the number of regressors in the unrestricted model excluding the intercept.

Under H_0 , the reference distribution is

$$F \sim F_{q, N-k-1} = F_{2, 527}.$$

Part 2

Exercise 6: Error Variance and OLS Precision

The relevant formulas are:

Under Homoscedasticity:

$$\text{Var}(\hat{\beta} \mid X) = \sigma^2(X'X)^{-1}$$

Under Heteroscedasticity:

$$\text{Var}(\hat{\beta} \mid X) = (X'X)^{-1}X'\Omega X(X'X)^{-1}$$

(a) Difference between the homo- and the heteroscedasticity formulas

Under homoscedasticity the formula assumes that the conditional error variance is the same for all observations, so one common scalar variance σ^2 is enough.

Under heteroscedasticity, the conditional error variance is not constant but can differ across observations, so the simple scalar σ^2 is replaced by the matrix Ω . $\Omega = \text{Var}(u | X)$ is the conditional variance-covariance matrix of the error terms, with the diagonal entries $\sigma_1^2, \dots, \sigma_N^2$, if errors are uncorrelated across observations, as seen in the following representation of the matrix:

$$\Omega = \begin{pmatrix} \sigma_1^2 & 0 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & 0 & \cdots & 0 \\ 0 & 0 & \sigma_3^2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \sigma_N^2 \end{pmatrix}$$

(b) Problems with the homoscedastic OLS standard error

As shown in Question 3, the residuals are more spread out for highly educated households, so the error variance appears to depend on education. The homoscedastic OLS standard error formula ignores this and treats all observations as having the same error variance. Therefore, the reported standard errors may be wrong, leading to misleading t -tests, p -values, and confidence intervals.

(c) Homoscedasticity: Ω

Under homoscedasticity, every observation has the same conditional error variance:

$$\text{Var}(u_i | X) = \sigma^2 \quad \text{for all } i.$$

Therefore, the matrix $\Omega = \text{Var}(u | X)$ has the same value σ^2 on every diagonal entry, and, assuming errors are uncorrelated across observations, zeros on the off-diagonal entries:

$$\Omega = \begin{pmatrix} \sigma^2 & 0 & \cdots & 0 \\ 0 & \sigma^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma^2 \end{pmatrix} = \sigma^2 I_N,$$

where I_N is the $N \times N$ identity matrix.

Substituting this into the heteroscedasticity formula gives

$$(X'X)^{-1}X'\Omega X(X'X)^{-1} = (X'X)^{-1}X'(\sigma^2 I_N)X(X'X)^{-1} = \sigma^2(X'X)^{-1}X'X(X'X)^{-1} = \sigma^2(X'X)^{-1}.$$

(d) “if OLS is unbiased, then the usual OLS standard errors are automatically valid?”

No, OLS can be unbiased if MLR.1 to MLR.4 hold, but the usual OLS standard errors are not automatically valid because they also require the correct variance assumption. Under heteroscedasticity, the coefficients may still be unbiased, but the homoscedastic standard errors, and therefore the resulting inference, can be wrong.